Geography of Colombian Life Satisfaction

Monday 26th October, 2026 20:30

ONLINE APPENDIX (Supplementary Online Material (SOM))

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead]

Contents

1 Literature–initial take **2**

2 Significance testing **3**

2.1 Rural and Farmer . . . 3

2.1.1 Rural . . . 3

2.1.2 Farmer . . . 6

3 Additional Ecological Predictors at Department Level **9**

3.1 PCGDP . . . 9

3.2 Population Density . . . 10

3.3 Crime . . . 11

4 Re-estimation Using 2021 ECV **12**

4.1 Departments/Provinces . . . 12

4.2 Cities/Municipalities . . . 14

4.3 Urban-Rural and Farmers . . . 17

5 Geography of Colombia **19**

5.1 Administrative Boundaries . . . 19

5.2 Physical Geography . . . 20

5.3 Socio-cultural Geography . . . 22

1 Literature–initial take

One of the puzzles in the happiness literature is Latin American phenomenon (Rojas 2015, 2019): high happiness[1] despite low development in Latin America. Colombia is one of the happiest countries in the world–see table 1. But we know little about geography of happiness within Colombia. This paper is about geography of Colombian happiness across its geographies: departments, municipalities, and farmers v non-farmers.

Table 1: 10 happiest countries in the world. Data from World Database of Happiness (WDH) 2010-2019 out of 160 countries at `worlddatabaseofhappiness.eur.nl/rank-reports/satisfaction-with-life`; and World Values Surveys (WVS) 2005-2022 [waves 5-7] out of 88 countries at `https://www.worldvaluessurvey.org`.

WDH			WVS		
rank	country	happiness (1-10)	rank	country	happiness (1-10)
1	Denmark	8.2	1	Puerto Rico	8.4
2	Mexico	8.1	2	Mexico	8.3
3	Colombia	8.1	3	Colombia	8.3
4	Switzerland	8	4	Qatar	8.0
5	Finland	8	5	Norway	7.9
6	Iceland	8	6	Nicaragua	7.9
7	Costa Rica	7.9	7	Tajikistan	7.9
8	Norway	7.9	8	Switzerland	7.9
9	Canada	7.9	9	Uzbekistan	7.9
10	Qatar	7.8	10	Ecuador	7.8

A Peruvian social psychologist specializing in happiness gives a persuasive explanation of (unexpectedly) high Latin happiness:

> the minimalist well-being lessons of Andean and Amazonian small traditional communities which constitute the grounds of Latin American happiness, a life style that mimics the ancestral environment, the deep nature where the happiness brain wiring occurred; a physical and social environment that naturally activates the brain pleasure circuits. Culture resemble evolutionary needs; resources to achieve needs are available for everyone; positive, interdependent collectivist interaction is ingrained in behavior, supporting, working, competing, and sharing. (Yamamoto 2016, p.45)

There is a Latin American concept of 'buen vivir' (good living) that is about living close to nature and other people, in balance and sustainably, rejecting extractive materialistic and consumerist capitalism (Hidalgo-Capitán and Cubillo-Guevara 2017, Calisto Friant and Langmore 2015, Guardiola and García-Quero 2014, García-Quero and Guardiola 2018). Guardiola and García-Quero (2014), García-Quero and Guardiola (2018) are probably the most relevant here and they are also next door, in Ecuador.

Nature and community contribute to SWB, not just employment income and consumption (Guardiola and García-Quero 2014)–something largely forgotten in the US (Okulicz-Kozaryn and Martinez 2025). For instance, households that grow their own food and are in an indigenous community depend less on money to be happy (García-Quero and Guardiola 2018). Then we would expect that high happiness in Colombia will be found not only in the richest urban parts of the country but also in natural communal rurality. There could be some urban buen vivir especially if city is not too large or dense and with good access to nature, and there could be no buen vivir in rural areas with industrial extractive agriculture and other problems. But in general probably the more rural, the more nature, community, and buen vivir.

We know very little about geography of Colombian happiness–there are only 4 studies on the topic as far as we can tell: Peiró-Palomino et al. (2020)/Peiro-Palomino et al. (2021), PNUD (2023), Burger et al. (2021), and Okulicz-Kozaryn et al. (2026).

Peiró-Palomino et al. (2020)/Peiro-Palomino et al. (2021)[2] is a pioneering study here, but it uses an index of objective indicators of quality of life such as income, housing and pollution. There is no subjective wellbeing studied here. Still, their study has a useful

[1] Following usual practice in social indicators research terms happiness and life satisfaction are used interchangeably, and specifically we mean cognitive and evaluative life satisfaction rather than affective happiness.

[2] Both are the same study.

information for our study, especially the map in their Figure 1: Global well-being in the Colombian departments, 2016. Their objective wellbeing findings are quite similar to our subjective wellbeing findings in next sections.

Burger et al. (2021) does analyze subjective wellbeing geography within Colombia, but has few survey observations: only several thousand (7-9k depending on model), and problematic measurement across space–it uses questionable Gallup data (see discussion in Okulicz-Kozaryn and Valente 2021), hence, we leave it out. Okulicz-Kozaryn et al. (2026) only focuses or urban-rural. While it also breaks down the analyses by municipalities, it only addresses Bogota, and lumps all the other municipalities into 2 categories (and like Burger et al. (2021) suffers from small sample).

Finally, the most comprehensive study to date: PNUD (2023) is a recent United Nations study on Colombian happiness using the same Encuesta Nacional de Calidad de Vida that we use, just earlier waves. The study is available from the the UN.[3] It is closest to our study and most useful overview of Colombian happiness geography to date. There is in fact a similar map to ours in Figure 1.9 on page 26 for 2018-2021. While happy Cafetero agrees with our analysis, happy Narino does not, and it finds Guaviare to be unhappy, while we find it moderately happy. A possible explanation for the differences is different time-frame: their 2018-2021 v our 2023. Likewise in figure 1.10 they break down SWB by urban-rural, our results agree that there is almost no difference, but we elaborate the relationship here: in addition to urban-rural we add farmer-nonfarmer and income breakdowns. PNUD (2023) also does analysis for municipalities in figures 2.14, 2.17 and 2.21 but the analyses do not address SWB, and only cover 10 cities, while we cover >40.

Overall, while there are at least 4 studies on the topic, only one (PNUD 2023) really has started addressing the life satisfaction geography of Colombia. These are, however, only rudimentary analyses among many, and ours is really the first study on the geography of Colombian life satisfaction.

2 Significance testing

2.1 Rural and Farmer

2.1.1 Rural

```
. reg life_satisfaction i.rural##ib10.inc10, robust
Linear regression                               Number of obs     =    184,047
                                                F(19, 184027)     =     256.08
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0255
                                                Root MSE          =     1.6726
------------------------------------------------------------------------------
             |               Robust
life_satis~n | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     1.rural |      -0.16       0.02    -7.19    0.00        -0.21       -0.12
             |
       inc10 |
          1  |      -0.90       0.03   -34.58    0.00        -0.95       -0.85
          2  |      -0.97       0.03   -34.09    0.00        -1.03       -0.91
          3  |      -0.91       0.03   -32.62    0.00        -0.96       -0.85
          4  |      -0.75       0.03   -28.79    0.00        -0.80       -0.70
          5  |      -0.61       0.02   -25.43    0.00        -0.66       -0.56
          6  |      -0.60       0.02   -27.19    0.00        -0.65       -0.56
          7  |      -0.47       0.02   -21.64    0.00        -0.51       -0.43
          8  |      -0.41       0.02   -20.60    0.00        -0.45       -0.37
          9  |      -0.26       0.02   -14.39    0.00        -0.30       -0.22
             |
 rural#inc10 |
       1  1  |       0.17       0.04     4.43    0.00         0.09        0.24
       1  2  |       0.27       0.04     6.83    0.00         0.19        0.34
       1  3  |       0.40       0.04    10.57    0.00         0.32        0.47
       1  4  |       0.35       0.04     9.65    0.00         0.28        0.42
       1  5  |       0.32       0.03     9.11    0.00         0.25        0.38
       1  6  |       0.32       0.03     9.70    0.00         0.26        0.39
       1  7  |       0.23       0.03     6.98    0.00         0.17        0.30
       1  8  |       0.27       0.03     8.34    0.00         0.20        0.33
       1  9  |       0.17       0.03     5.36    0.00         0.11        0.23
             |
       _cons |       8.56       0.01   739.21    0.00         8.54        8.58
------------------------------------------------------------------------------

BELOW JUST SELECTED CONTRASTS, ALL AFTER THAT
. margins rural, at(inc10=(1(1)10)) pwcompare(effects)
. margins rural, at(inc10=(1(1)10)) pwcompare(effects)
```

[3] https://www.undp.org/sites/g/files/zskgke326/files/2023-02/undp_co_pub_Percepciones_bienestar_subjetivo_2do_cuaderno_INDH_Feb21_2023.pdf.

```
Pairwise comparisons of adjusted predictions           Number of obs = 184,047
Model VCE: Robust

Expression: Linear prediction, predict()
1._at:  inc10 =  1
2._at:  inc10 =  2
3._at:  inc10 =  3
4._at:  inc10 =  4
5._at:  inc10 =  5
6._at:  inc10 =  6
7._at:  inc10 =  7
8._at:  inc10 =  8
9._at:  inc10 =  9
10._at: inc10 = 10
------------------------------------------------------------------------------
                  |            Delta-method    Unadjusted           Unadjusted
                  |   Contrast   std. err.      t    P>|t|     [95% conf. interval]
------------------+-----------------------------------------------------------
        _at#rural |
( 1 1) vs ( 1 0)  |       0.00       0.03     0.13    0.90       -0.05        0.06
insig
( 2 1) vs ( 2 0)  |       0.10       0.03     3.26    0.00        0.04        0.17
sig
( 3 1) vs ( 3 0)  |       0.24       0.03     7.82    0.00        0.18        0.29
sig
( 4 1) vs ( 4 0)  |       0.19       0.03     6.61    0.00        0.13        0.24
sig
( 5 1) vs ( 5 0)  |       0.15       0.03     5.83    0.00        0.10        0.21
sig
( 6 1) vs ( 6 0)  |       0.16       0.02     6.55    0.00        0.11        0.21
sig
( 7 1) vs ( 7 0)  |       0.07       0.02     2.83    0.00        0.02        0.12
sig
( 8 1) vs ( 8 0)  |       0.10       0.02     4.59    0.00        0.06        0.15
sig
( 9 1) vs ( 9 0)  |       0.00       0.02     0.22    0.83       -0.04        0.05
insig
(10 1) vs (10 0)  |      -0.16       0.02    -7.19    0.00       -0.21       -0.12
sig



. margins rural, at(inc10=(1(1)10)) pwcompare(effects)
Pairwise comparisons of adjusted predictions           Number of obs = 184,047
Model VCE: Robust

Expression: Linear prediction, predict()
1._at:  inc10 =  1
2._at:  inc10 =  2
3._at:  inc10 =  3
4._at:  inc10 =  4
5._at:  inc10 =  5
6._at:  inc10 =  6
7._at:  inc10 =  7
8._at:  inc10 =  8
9._at:  inc10 =  9
10._at: inc10 = 10
------------------------------------------------------------------------------
                  |            Delta-method    Unadjusted           Unadjusted
                  |   Contrast   std. err.      t    P>|t|     [95% conf. interval]
------------------+-----------------------------------------------------------
        _at#rural |
( 1 1) vs ( 1 0)  |       0.00       0.03     0.13    0.90       -0.05        0.06
( 2 0) vs ( 1 0)  |      -0.07       0.03    -2.03    0.04       -0.14       -0.00
( 2 1) vs ( 1 0)  |       0.03       0.03     1.10    0.27       -0.03        0.09
( 3 0) vs ( 1 0)  |      -0.01       0.03    -0.27    0.78       -0.08        0.06
( 3 1) vs ( 1 0)  |       0.23       0.03     7.96    0.00        0.17        0.28
( 4 0) vs ( 1 0)  |       0.15       0.03     4.59    0.00        0.09        0.22
( 4 1) vs ( 1 0)  |       0.34       0.03    11.96    0.00        0.28        0.39
( 5 0) vs ( 1 0)  |       0.29       0.03     9.15    0.00        0.23        0.35
( 5 1) vs ( 1 0)  |       0.44       0.03    15.68    0.00        0.39        0.50
( 6 0) vs ( 1 0)  |       0.30       0.03     9.83    0.00        0.24        0.35
( 6 1) vs ( 1 0)  |       0.45       0.03    16.37    0.00        0.40        0.51
( 7 0) vs ( 1 0)  |       0.43       0.03    14.56    0.00        0.37        0.49
( 7 1) vs ( 1 0)  |       0.50       0.03    17.76    0.00        0.44        0.55
( 8 0) vs ( 1 0)  |       0.49       0.03    17.21    0.00        0.43        0.54
( 8 1) vs ( 1 0)  |       0.59       0.03    21.12    0.00        0.54        0.65
( 9 0) vs ( 1 0)  |       0.64       0.03    23.56    0.00        0.59        0.69
( 9 1) vs ( 1 0)  |       0.64       0.03    22.57    0.00        0.59        0.70
(10 0) vs ( 1 0)  |       0.90       0.03    34.58    0.00        0.85        0.95
(10 1) vs ( 1 0)  |       0.74       0.03    24.27    0.00        0.68        0.80
( 2 0) vs ( 1 1)  |      -0.07       0.03    -2.32    0.02       -0.14       -0.01
( 2 1) vs ( 1 1)  |       0.03       0.03     1.09    0.28       -0.02        0.08
( 3 0) vs ( 1 1)  |      -0.01       0.03    -0.42    0.68       -0.08        0.05
( 3 1) vs ( 1 1)  |       0.22       0.02     8.92    0.00        0.17        0.27
( 4 0) vs ( 1 1)  |       0.15       0.03     4.91    0.00        0.09        0.21
( 4 1) vs ( 1 1)  |       0.33       0.02    13.51    0.00        0.29        0.38
( 5 0) vs ( 1 1)  |       0.28       0.03    10.02    0.00        0.23        0.34
( 5 1) vs ( 1 1)  |       0.44       0.02    17.77    0.00        0.39        0.49
( 6 0) vs ( 1 1)  |       0.29       0.03    10.88    0.00        0.24        0.34
( 6 1) vs ( 1 1)  |       0.45       0.02    18.63    0.00        0.40        0.50
( 7 0) vs ( 1 1)  |       0.43       0.03    16.26    0.00        0.38        0.48
( 7 1) vs ( 1 1)  |       0.50       0.02    20.15    0.00        0.45        0.54
( 8 0) vs ( 1 1)  |       0.48       0.02    19.46    0.00        0.44        0.53
( 8 1) vs ( 1 1)  |       0.59       0.02    24.00    0.00        0.54        0.64
( 9 0) vs ( 1 1)  |       0.64       0.02    27.08    0.00        0.59        0.68
( 9 1) vs ( 1 1)  |       0.64       0.03    25.53    0.00        0.59        0.69
(10 0) vs ( 1 1)  |       0.90       0.02    40.42    0.00        0.85        0.94
(10 1) vs ( 1 1)  |       0.73       0.03    27.02    0.00        0.68        0.79
( 2 1) vs ( 2 0)  |       0.10       0.03     3.26    0.00        0.04        0.17
( 3 0) vs ( 2 0)  |       0.06       0.04     1.69    0.09       -0.01        0.13
( 3 1) vs ( 2 0)  |       0.30       0.03     9.68    0.00        0.24        0.36
( 4 0) vs ( 2 0)  |       0.22       0.03     6.36    0.00        0.15        0.29
( 4 1) vs ( 2 0)  |       0.41       0.03    13.39    0.00        0.35        0.47
```

(5 0) vs (2 0)	0.36	0.03	10.70	0.00	0.29	0.42
(5 1) vs (2 0)	0.51	0.03	16.83	0.00	0.45	0.57
(6 0) vs (2 0)	0.37	0.03	11.37	0.00	0.30	0.43
(6 1) vs (2 0)	0.52	0.03	17.47	0.00	0.47	0.58
(7 0) vs (2 0)	0.50	0.03	15.79	0.00	0.44	0.56
(7 1) vs (2 0)	0.57	0.03	18.76	0.00	0.51	0.63
(8 0) vs (2 0)	0.56	0.03	18.25	0.00	0.50	0.62
(8 1) vs (2 0)	0.66	0.03	21.86	0.00	0.60	0.72
(9 0) vs (2 0)	0.71	0.03	24.08	0.00	0.65	0.77
(9 1) vs (2 0)	0.71	0.03	23.22	0.00	0.65	0.78
(10 0) vs (2 0)	0.97	0.03	34.09	0.00	0.91	1.03
(10 1) vs (2 0)	0.81	0.03	24.86	0.00	0.74	0.87
(3 0) vs (2 1)	-0.04	0.03	-1.34	0.18	-0.10	0.02
(3 1) vs (2 1)	0.19	0.02	7.97	0.00	0.15	0.24
(4 0) vs (2 1)	0.12	0.03	4.03	0.00	0.06	0.18
(4 1) vs (2 1)	0.31	0.02	12.68	0.00	0.26	0.35
(5 0) vs (2 1)	0.26	0.03	9.19	0.00	0.20	0.31
(5 1) vs (2 1)	0.41	0.02	17.04	0.00	0.36	0.46
(6 0) vs (2 1)	0.26	0.03	10.04	0.00	0.21	0.31
(6 1) vs (2 1)	0.42	0.02	17.93	0.00	0.38	0.47
(7 0) vs (2 1)	0.40	0.03	15.52	0.00	0.35	0.45
(7 1) vs (2 1)	0.47	0.02	19.49	0.00	0.42	0.51
(8 0) vs (2 1)	0.46	0.02	18.79	0.00	0.41	0.50
(8 1) vs (2 1)	0.56	0.02	23.44	0.00	0.51	0.61
(9 0) vs (2 1)	0.61	0.02	26.62	0.00	0.56	0.65
(9 1) vs (2 1)	0.61	0.02	25.01	0.00	0.56	0.66
(10 0) vs (2 1)	0.87	0.02	40.42	0.00	0.83	0.91
(10 1) vs (2 1)	0.70	0.03	26.52	0.00	0.65	0.76
(3 1) vs (3 0)	0.24	0.03	7.82	0.00	0.18	0.29
(4 0) vs (3 0)	0.16	0.03	4.66	0.00	0.09	0.23
(4 1) vs (3 0)	0.35	0.03	11.59	0.00	0.29	0.41
(5 0) vs (3 0)	0.30	0.03	9.00	0.00	0.23	0.36
(5 1) vs (3 0)	0.45	0.03	15.09	0.00	0.39	0.51
(6 0) vs (3 0)	0.30	0.03	9.62	0.00	0.24	0.37
(6 1) vs (3 0)	0.46	0.03	15.72	0.00	0.41	0.52
(7 0) vs (3 0)	0.44	0.03	14.10	0.00	0.38	0.50
(7 1) vs (3 0)	0.51	0.03	17.05	0.00	0.45	0.57
(8 0) vs (3 0)	0.50	0.03	16.55	0.00	0.44	0.56
(8 1) vs (3 0)	0.60	0.03	20.21	0.00	0.54	0.66
(9 0) vs (3 0)	0.65	0.03	22.44	0.00	0.59	0.71
(9 1) vs (3 0)	0.65	0.03	21.61	0.00	0.59	0.71
(10 0) vs (3 0)	0.91	0.03	32.62	0.00	0.85	0.96
(10 1) vs (3 0)	0.75	0.03	23.35	0.00	0.68	0.81
(4 0) vs (3 1)	-0.07	0.03	-2.64	0.01	-0.13	-0.02
(4 1) vs (3 1)	0.11	0.02	4.91	0.00	0.07	0.16
(5 0) vs (3 1)	0.06	0.03	2.32	0.02	0.01	0.11
(5 1) vs (3 1)	0.22	0.02	9.52	0.00	0.17	0.26
(6 0) vs (3 1)	0.07	0.02	2.78	0.01	0.02	0.12
(6 1) vs (3 1)	0.23	0.02	10.32	0.00	0.18	0.27
(7 0) vs (3 1)	0.21	0.02	8.41	0.00	0.16	0.25
(7 1) vs (3 1)	0.27	0.02	12.11	0.00	0.23	0.32
(8 0) vs (3 1)	0.26	0.02	11.46	0.00	0.22	0.31
(8 1) vs (3 1)	0.37	0.02	16.28	0.00	0.32	0.41
(9 0) vs (3 1)	0.41	0.02	19.36	0.00	0.37	0.46
(9 1) vs (3 1)	0.42	0.02	18.09	0.00	0.37	0.46
(10 0) vs (3 1)	0.67	0.02	33.83	0.00	0.63	0.71
(10 1) vs (3 1)	0.51	0.03	20.17	0.00	0.46	0.56
(4 1) vs (4 0)	0.19	0.03	6.61	0.00	0.13	0.24
(5 0) vs (4 0)	0.14	0.03	4.34	0.00	0.07	0.20
(5 1) vs (4 0)	0.29	0.03	10.31	0.00	0.23	0.35
(6 0) vs (4 0)	0.14	0.03	4.80	0.00	0.09	0.20
(6 1) vs (4 0)	0.30	0.03	10.93	0.00	0.25	0.36
(7 0) vs (4 0)	0.28	0.03	9.46	0.00	0.22	0.34
(7 1) vs (4 0)	0.35	0.03	12.39	0.00	0.29	0.40
(8 0) vs (4 0)	0.34	0.03	11.89	0.00	0.28	0.39
(8 1) vs (4 0)	0.44	0.03	15.74	0.00	0.39	0.50
(9 0) vs (4 0)	0.49	0.03	18.00	0.00	0.44	0.54
(9 1) vs (4 0)	0.49	0.03	17.28	0.00	0.44	0.55
(10 0) vs (4 0)	0.75	0.03	28.79	0.00	0.70	0.80
(10 1) vs (4 0)	0.59	0.03	19.30	0.00	0.53	0.65
(5 0) vs (4 1)	-0.05	0.03	-1.88	0.06	-0.10	0.00
(5 1) vs (4 1)	0.10	0.02	4.64	0.00	0.06	0.15
(6 0) vs (4 1)	-0.04	0.02	-1.70	0.09	-0.09	0.01
(6 1) vs (4 1)	0.12	0.02	5.34	0.00	0.07	0.16
(7 0) vs (4 1)	0.09	0.02	3.88	0.00	0.05	0.14
(7 1) vs (4 1)	0.16	0.02	7.25	0.00	0.12	0.21
(8 0) vs (4 1)	0.15	0.02	6.67	0.00	0.11	0.20
(8 1) vs (4 1)	0.25	0.02	11.45	0.00	0.21	0.30
(9 0) vs (4 1)	0.30	0.02	14.31	0.00	0.26	0.34
(9 1) vs (4 1)	0.31	0.02	13.40	0.00	0.26	0.35
(10 0) vs (4 1)	0.56	0.02	28.61	0.00	0.52	0.60
(10 1) vs (4 1)	0.40	0.03	15.90	0.00	0.35	0.45
(5 1) vs (5 0)	0.15	0.03	5.83	0.00	0.10	0.21
(6 0) vs (5 0)	0.01	0.03	0.27	0.78	-0.05	0.06
(6 1) vs (5 0)	0.17	0.03	6.42	0.00	0.12	0.22
(7 0) vs (5 0)	0.14	0.03	5.15	0.00	0.09	0.20
(7 1) vs (5 0)	0.21	0.03	8.05	0.00	0.16	0.26
(8 0) vs (5 0)	0.20	0.03	7.56	0.00	0.15	0.25
(8 1) vs (5 0)	0.30	0.03	11.61	0.00	0.25	0.36
(9 0) vs (5 0)	0.35	0.03	13.92	0.00	0.30	0.40
(9 1) vs (5 0)	0.36	0.03	13.31	0.00	0.30	0.41
(10 0) vs (5 0)	0.61	0.02	25.43	0.00	0.56	0.66
(10 1) vs (5 0)	0.45	0.03	15.65	0.00	0.39	0.51
(6 0) vs (5 1)	-0.15	0.02	-5.91	0.00	-0.19	-0.10
(6 1) vs (5 1)	0.01	0.02	0.59	0.55	-0.03	0.06
(7 0) vs (5 1)	-0.01	0.02	-0.41	0.68	-0.06	0.04
(7 1) vs (5 1)	0.06	0.02	2.62	0.01	0.01	0.10
(8 0) vs (5 1)	0.05	0.02	2.10	0.04	0.00	0.09
(8 1) vs (5 1)	0.15	0.02	6.80	0.00	0.11	0.19
(9 0) vs (5 1)	0.20	0.02	9.42	0.00	0.16	0.24
(9 1) vs (5 1)	0.20	0.02	8.89	0.00	0.16	0.25
(10 0) vs (5 1)	0.46	0.02	23.42	0.00	0.42	0.50
(10 1) vs (5 1)	0.30	0.03	11.80	0.00	0.25	0.34
(6 1) vs (6 0)	0.16	0.02	6.55	0.00	0.11	0.21
(7 0) vs (6 0)	0.14	0.03	5.16	0.00	0.08	0.19
(7 1) vs (6 0)	0.20	0.02	8.28	0.00	0.16	0.25
(8 0) vs (6 0)	0.19	0.02	7.74	0.00	0.14	0.24

```
( 8 1) vs ( 6 0)  |       0.30       0.02    12.08    0.00         0.25        0.35
( 9 0) vs ( 6 0)  |       0.34       0.02    14.62    0.00         0.30        0.39
( 9 1) vs ( 6 0)  |       0.35       0.03    13.87    0.00         0.30        0.40
(10 0) vs ( 6 0)  |       0.60       0.02    27.19    0.00         0.56        0.65
(10 1) vs ( 6 0)  |       0.44       0.03    16.24    0.00         0.39        0.49
( 7 0) vs ( 6 1)  |      -0.02       0.02    -0.96    0.34        -0.07        0.02
( 7 1) vs ( 6 1)  |       0.05       0.02     2.08    0.04         0.00        0.09
( 8 0) vs ( 6 1)  |       0.03       0.02     1.56    0.12        -0.01        0.08
( 8 1) vs ( 6 1)  |       0.14       0.02     6.36    0.00         0.10        0.18
( 9 0) vs ( 6 1)  |       0.19       0.02     9.03    0.00         0.15        0.23
( 9 1) vs ( 6 1)  |       0.19       0.02     8.50    0.00         0.15        0.23
(10 0) vs ( 6 1)  |       0.45       0.02    23.45    0.00         0.41        0.48
(10 1) vs ( 6 1)  |       0.28       0.02    11.48    0.00         0.23        0.33
( 7 1) vs ( 7 0)  |       0.07       0.02     2.83    0.00         0.02        0.12
( 8 0) vs ( 7 0)  |       0.06       0.02     2.35    0.02         0.01        0.11
( 8 1) vs ( 7 0)  |       0.16       0.02     6.69    0.00         0.11        0.21
( 9 0) vs ( 7 0)  |       0.21       0.02     9.05    0.00         0.16        0.25
( 9 1) vs ( 7 0)  |       0.21       0.02     8.64    0.00         0.16        0.26
(10 0) vs ( 7 0)  |       0.47       0.02    21.64    0.00         0.43        0.51
(10 1) vs ( 7 0)  |       0.31       0.03    11.44    0.00         0.25        0.36
( 8 0) vs ( 7 1)  |      -0.01       0.02    -0.48    0.63        -0.06        0.03
( 8 1) vs ( 7 1)  |       0.09       0.02     4.18    0.00         0.05        0.14
( 9 0) vs ( 7 1)  |       0.14       0.02     6.65    0.00         0.10        0.18
( 9 1) vs ( 7 1)  |       0.14       0.02     6.34    0.00         0.10        0.19
(10 0) vs ( 7 1)  |       0.40       0.02    20.47    0.00         0.36        0.44
(10 1) vs ( 7 1)  |       0.24       0.03     9.47    0.00         0.19        0.29
( 8 1) vs ( 8 0)  |       0.10       0.02     4.59    0.00         0.06        0.15
( 9 0) vs ( 8 0)  |       0.15       0.02     7.04    0.00         0.11        0.19
( 9 1) vs ( 8 0)  |       0.16       0.02     6.71    0.00         0.11        0.20
(10 0) vs ( 8 0)  |       0.41       0.02    20.60    0.00         0.37        0.45
(10 1) vs ( 8 0)  |       0.25       0.03     9.78    0.00         0.20        0.30
( 9 0) vs ( 8 1)  |       0.05       0.02     2.25    0.02         0.01        0.09
( 9 1) vs ( 8 1)  |       0.05       0.02     2.28    0.02         0.01        0.10
(10 0) vs ( 8 1)  |       0.31       0.02    15.81    0.00         0.27        0.35
(10 1) vs ( 8 1)  |       0.14       0.02     5.79    0.00         0.10        0.19
( 9 1) vs ( 9 0)  |       0.00       0.02     0.22    0.83        -0.04        0.05
(10 0) vs ( 9 0)  |       0.26       0.02    14.39    0.00         0.22        0.30
(10 1) vs ( 9 0)  |       0.10       0.02     4.07    0.00         0.05        0.14
(10 0) vs ( 9 1)  |       0.26       0.02    12.69    0.00         0.22        0.30
(10 1) vs ( 9 1)  |       0.09       0.03     3.64    0.00         0.04        0.14
(10 1) vs (10 0)  |      -0.16       0.02    -7.19    0.00        -0.21       -0.12
------------------------------------------------------------------------------
```

2.1.2 Farmer

```
. reg life_satisfaction i.farmer#ib10.inc10, robust
Linear regression                               Number of obs     =    185,236
                                                F(19, 185216)     =     255.75
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0250
                                                Root MSE          =     1.6737
------------------------------------------------------------------------------
             |               Robust
life_satis~n | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
farmer#inc10 |
       0  1  |      -0.90       0.03   -34.92    0.00        -0.95       -0.85
       0  2  |      -0.95       0.03   -34.41    0.00        -1.00       -0.89
       0  3  |      -0.87       0.03   -32.33    0.00        -0.93       -0.82
       0  4  |      -0.72       0.03   -28.14    0.00        -0.77       -0.67
       0  5  |      -0.60       0.02   -25.34    0.00        -0.65       -0.55
       0  6  |      -0.57       0.02   -25.88    0.00        -0.62       -0.53
       0  7  |      -0.46       0.02   -21.40    0.00        -0.50       -0.42
       0  8  |      -0.39       0.02   -19.73    0.00        -0.43       -0.35
       0  9  |      -0.26       0.02   -14.27    0.00        -0.29       -0.22
       1  1  |      -0.91       0.02   -41.12    0.00        -0.96       -0.87
       1  2  |      -0.88       0.02   -40.86    0.00        -0.93       -0.84
       1  3  |      -0.70       0.02   -34.70    0.00        -0.73       -0.66
       1  4  |      -0.58       0.02   -29.65    0.00        -0.62       -0.55
       1  5  |      -0.47       0.02   -24.14    0.00        -0.51       -0.44
       1  6  |      -0.48       0.02   -25.30    0.00        -0.52       -0.44
       1  7  |      -0.42       0.02   -21.32    0.00        -0.46       -0.38
       1  8  |      -0.34       0.02   -17.49    0.00        -0.38       -0.30
       1  9  |      -0.27       0.02   -13.48    0.00        -0.31       -0.23
       1 10  |      -0.17       0.02    -7.59    0.00        -0.22       -0.13
             |
       _cons |       8.56       0.01   745.87    0.00         8.54        8.59
------------------------------------------------------------------------------


BELOW JUST SELECTED CONTRASTS, ALL AFTER THAT
margins farmer, at(inc10=(1(1)10)) pwcompare(effects)
Pairwise comparisons of adjusted predictions           Number of obs = 185,236
Model VCE: Robust

Expression: Linear prediction, predict()
1._at:  inc10 =  1
2._at:  inc10 =  2
3._at:  inc10 =  3
4._at:  inc10 =  4
5._at:  inc10 =  5
6._at:  inc10 =  6
7._at:  inc10 =  7
8._at:  inc10 =  8
9._at:  inc10 =  9
10._at: inc10 = 10
------------------------------------------------------------------------------
                 |            Delta-method    Unadjusted           Unadjusted
                 |   Contrast   std. err.      t    P>|t|     [95% conf. interval]
-----------------+------------------------------------------------------------
      _at#farmer |
( 1 1) vs ( 1 0)  |      -0.02       0.03    -0.60    0.55        -0.08        0.04
```

```
insig
( 2 1) vs ( 2 0)  |       0.06       0.03     2.03    0.04         0.00        0.12
sig
( 3 1) vs ( 3 0)  |       0.18       0.03     6.07    0.00         0.12        0.24
sig
( 4 1) vs ( 4 0)  |       0.13       0.03     4.77    0.00         0.08        0.19
sig
( 5 1) vs ( 5 0)  |       0.13       0.03     4.86    0.00         0.08        0.18
sig
( 6 1) vs ( 6 0)  |       0.10       0.02     3.98    0.00         0.05        0.14
sig
( 7 1) vs ( 7 0)  |       0.04       0.02     1.71    0.09        -0.01        0.09
insig; only sig at .1
( 8 1) vs ( 8 0)  |       0.05       0.02     2.41    0.02         0.01        0.10
sig

( 9 1) vs ( 9 0)  |      -0.01       0.02    -0.48    0.63        -0.05        0.03
insig
(10 1) vs (10 0)  |      -0.17       0.02    -7.59    0.00        -0.22       -0.13
sig
```

```
. margins farmer, at(inc10=(1(1)10)) pwcompare(effects)
margins farmer, at(inc10=(1(1)10)) pwcompare(effects)
Pairwise comparisons of adjusted predictions          Number of obs = 185,236
Model VCE: Robust

Expression: Linear prediction, predict()
1._at:  inc10 =  1
2._at:  inc10 =  2
3._at:  inc10 =  3
4._at:  inc10 =  4
5._at:  inc10 =  5
6._at:  inc10 =  6
7._at:  inc10 =  7
8._at:  inc10 =  8
9._at:  inc10 =  9
10._at: inc10 = 10
------------------------------------------------------------------------------
                  |            Delta-method    Unadjusted           Unadjusted
                  |   Contrast   std. err.      t    P>|t|     [95% conf. interval]
------------------+-----------------------------------------------------------
       _at#farmer |
( 1 1) vs ( 1 0)  |      -0.02       0.03    -0.60    0.55        -0.08        0.04
( 2 0) vs ( 1 0)  |      -0.05       0.03    -1.49    0.14        -0.12        0.02
( 2 1) vs ( 1 0)  |       0.01       0.03     0.42    0.68        -0.05        0.07
( 3 0) vs ( 1 0)  |       0.02       0.03     0.65    0.51        -0.04        0.09
( 3 1) vs ( 1 0)  |       0.20       0.03     7.13    0.00         0.15        0.26
( 4 0) vs ( 1 0)  |       0.18       0.03     5.52    0.00         0.12        0.24
( 4 1) vs ( 1 0)  |       0.31       0.03    11.12    0.00         0.26        0.37
( 5 0) vs ( 1 0)  |       0.30       0.03     9.55    0.00         0.23        0.36
( 5 1) vs ( 1 0)  |       0.42       0.03    15.12    0.00         0.37        0.48
( 6 0) vs ( 1 0)  |       0.32       0.03    10.78    0.00         0.26        0.38
( 6 1) vs ( 1 0)  |       0.42       0.03    15.23    0.00         0.36        0.47
( 7 0) vs ( 1 0)  |       0.44       0.03    14.98    0.00         0.38        0.50
( 7 1) vs ( 1 0)  |       0.48       0.03    17.16    0.00         0.42        0.53
( 8 0) vs ( 1 0)  |       0.50       0.03    17.90    0.00         0.45        0.56
( 8 1) vs ( 1 0)  |       0.56       0.03    20.11    0.00         0.50        0.61
( 9 0) vs ( 1 0)  |       0.64       0.03    23.73    0.00         0.59        0.69
( 9 1) vs ( 1 0)  |       0.63       0.03    22.30    0.00         0.57        0.68
(10 0) vs ( 1 0)  |       0.90       0.03    34.92    0.00         0.85        0.95
(10 1) vs ( 1 0)  |       0.72       0.03    24.05    0.00         0.67        0.78
( 2 0) vs ( 1 1)  |      -0.03       0.03    -1.04    0.30        -0.09        0.03
( 2 1) vs ( 1 1)  |       0.03       0.03     1.14    0.25        -0.02        0.08
( 3 0) vs ( 1 1)  |       0.04       0.03     1.29    0.20        -0.02        0.10
( 3 1) vs ( 1 1)  |       0.22       0.03     8.71    0.00         0.17        0.27
( 4 0) vs ( 1 1)  |       0.20       0.03     6.62    0.00         0.14        0.25
( 4 1) vs ( 1 1)  |       0.33       0.02    13.23    0.00         0.28        0.38
( 5 0) vs ( 1 1)  |       0.31       0.03    11.13    0.00         0.26        0.37
( 5 1) vs ( 1 1)  |       0.44       0.02    17.74    0.00         0.39        0.49
( 6 0) vs ( 1 1)  |       0.34       0.03    12.61    0.00         0.29        0.39
( 6 1) vs ( 1 1)  |       0.44       0.02    17.96    0.00         0.39        0.48
( 7 0) vs ( 1 1)  |       0.46       0.03    17.35    0.00         0.40        0.51
( 7 1) vs ( 1 1)  |       0.50       0.02    20.05    0.00         0.45        0.55
( 8 0) vs ( 1 1)  |       0.52       0.03    20.83    0.00         0.47        0.57
( 8 1) vs ( 1 1)  |       0.58       0.02    23.40    0.00         0.53        0.62
( 9 0) vs ( 1 1)  |       0.66       0.02    27.76    0.00         0.61        0.70
( 9 1) vs ( 1 1)  |       0.65       0.03    25.76    0.00         0.60        0.69
(10 0) vs ( 1 1)  |       0.91       0.02    41.12    0.00         0.87        0.96
(10 1) vs ( 1 1)  |       0.74       0.03    27.23    0.00         0.69        0.80
( 2 1) vs ( 2 0)  |       0.06       0.03     2.03    0.04         0.00        0.12
( 3 0) vs ( 2 0)  |       0.07       0.04     2.08    0.04         0.00        0.14
( 3 1) vs ( 2 0)  |       0.25       0.03     8.42    0.00         0.19        0.31
( 4 0) vs ( 2 0)  |       0.23       0.03     6.78    0.00         0.16        0.30
( 4 1) vs ( 2 0)  |       0.36       0.03    12.19    0.00         0.30        0.42
( 5 0) vs ( 2 0)  |       0.35       0.03    10.65    0.00         0.28        0.41
( 5 1) vs ( 2 0)  |       0.47       0.03    15.96    0.00         0.42        0.53
( 6 0) vs ( 2 0)  |       0.37       0.03    11.84    0.00         0.31        0.43
( 6 1) vs ( 2 0)  |       0.47       0.03    16.06    0.00         0.41        0.53
( 7 0) vs ( 2 0)  |       0.49       0.03    15.83    0.00         0.43        0.55
( 7 1) vs ( 2 0)  |       0.53       0.03    17.88    0.00         0.47        0.59
( 8 0) vs ( 2 0)  |       0.55       0.03    18.58    0.00         0.50        0.61
( 8 1) vs ( 2 0)  |       0.61       0.03    20.65    0.00         0.55        0.67
( 9 0) vs ( 2 0)  |       0.69       0.03    24.03    0.00         0.63        0.74
( 9 1) vs ( 2 0)  |       0.68       0.03    22.72    0.00         0.62        0.74
(10 0) vs ( 2 0)  |       0.95       0.03    34.41    0.00         0.89        1.00
(10 1) vs ( 2 0)  |       0.78       0.03    24.44    0.00         0.71        0.84
( 3 0) vs ( 2 1)  |       0.01       0.03     0.32    0.75        -0.05        0.07
```

(3 1) vs (2 1)	0.19	0.02	7.67	0.00	0.14	0.24
(4 0) vs (2 1)	0.17	0.03	5.69	0.00	0.11	0.22
(4 1) vs (2 1)	0.30	0.02	12.28	0.00	0.25	0.35
(5 0) vs (2 1)	0.28	0.03	10.23	0.00	0.23	0.34
(5 1) vs (2 1)	0.41	0.02	16.89	0.00	0.36	0.46
(6 0) vs (2 1)	0.31	0.03	11.70	0.00	0.26	0.36
(6 1) vs (2 1)	0.41	0.02	17.10	0.00	0.36	0.45
(7 0) vs (2 1)	0.43	0.03	16.52	0.00	0.38	0.48
(7 1) vs (2 1)	0.47	0.02	19.24	0.00	0.42	0.51
(8 0) vs (2 1)	0.49	0.02	20.05	0.00	0.44	0.54
(8 1) vs (2 1)	0.55	0.02	22.67	0.00	0.50	0.59
(9 0) vs (2 1)	0.63	0.02	27.12	0.00	0.58	0.67
(9 1) vs (2 1)	0.62	0.02	25.08	0.00	0.57	0.66
(10 0) vs (2 1)	0.88	0.02	40.86	0.00	0.84	0.93
(10 1) vs (2 1)	0.71	0.03	26.59	0.00	0.66	0.76
(3 1) vs (3 0)	0.18	0.03	6.07	0.00	0.12	0.24
(4 0) vs (3 0)	0.16	0.03	4.68	0.00	0.09	0.22
(4 1) vs (3 0)	0.29	0.03	9.89	0.00	0.23	0.35
(5 0) vs (3 0)	0.27	0.03	8.52	0.00	0.21	0.34
(5 1) vs (3 0)	0.40	0.03	13.71	0.00	0.34	0.46
(6 0) vs (3 0)	0.30	0.03	9.65	0.00	0.24	0.36
(6 1) vs (3 0)	0.40	0.03	13.77	0.00	0.34	0.45
(7 0) vs (3 0)	0.42	0.03	13.66	0.00	0.36	0.48
(7 1) vs (3 0)	0.46	0.03	15.66	0.00	0.40	0.51
(8 0) vs (3 0)	0.48	0.03	16.38	0.00	0.42	0.54
(8 1) vs (3 0)	0.54	0.03	18.46	0.00	0.48	0.59
(9 0) vs (3 0)	0.62	0.03	21.84	0.00	0.56	0.67
(9 1) vs (3 0)	0.61	0.03	20.59	0.00	0.55	0.66
(10 0) vs (3 0)	0.87	0.03	32.33	0.00	0.82	0.93
(10 1) vs (3 0)	0.70	0.03	22.43	0.00	0.64	0.76
(4 0) vs (3 1)	-0.02	0.03	-0.80	0.42	-0.08	0.03
(4 1) vs (3 1)	0.11	0.02	4.81	0.00	0.07	0.16
(5 0) vs (3 1)	0.09	0.03	3.57	0.00	0.04	0.15
(5 1) vs (3 1)	0.22	0.02	9.68	0.00	0.18	0.27
(6 0) vs (3 1)	0.12	0.03	4.79	0.00	0.07	0.17
(6 1) vs (3 1)	0.22	0.02	9.74	0.00	0.17	0.26
(7 0) vs (3 1)	0.24	0.02	9.69	0.00	0.19	0.28
(7 1) vs (3 1)	0.28	0.02	12.17	0.00	0.23	0.32
(8 0) vs (3 1)	0.30	0.02	13.09	0.00	0.26	0.35
(8 1) vs (3 1)	0.36	0.02	15.76	0.00	0.31	0.40
(9 0) vs (3 1)	0.44	0.02	20.25	0.00	0.39	0.48
(9 1) vs (3 1)	0.43	0.02	18.44	0.00	0.38	0.47
(10 0) vs (3 1)	0.70	0.02	34.70	0.00	0.66	0.73
(10 1) vs (3 1)	0.52	0.03	20.52	0.00	0.47	0.57
(4 1) vs (4 0)	0.13	0.03	4.77	0.00	0.08	0.19
(5 0) vs (4 0)	0.12	0.03	3.79	0.00	0.06	0.18
(5 1) vs (4 0)	0.24	0.03	8.77	0.00	0.19	0.30
(6 0) vs (4 0)	0.14	0.03	4.81	0.00	0.08	0.20
(6 1) vs (4 0)	0.24	0.03	8.77	0.00	0.19	0.29
(7 0) vs (4 0)	0.26	0.03	8.91	0.00	0.20	0.32
(7 1) vs (4 0)	0.30	0.03	10.82	0.00	0.25	0.35
(8 0) vs (4 0)	0.33	0.03	11.62	0.00	0.27	0.38
(8 1) vs (4 0)	0.38	0.03	13.74	0.00	0.33	0.43
(9 0) vs (4 0)	0.46	0.03	17.18	0.00	0.41	0.51
(9 1) vs (4 0)	0.45	0.03	16.03	0.00	0.39	0.50
(10 0) vs (4 0)	0.72	0.03	28.14	0.00	0.67	0.77
(10 1) vs (4 0)	0.55	0.03	18.20	0.00	0.49	0.60
(5 0) vs (4 1)	-0.02	0.03	-0.61	0.54	-0.07	0.04
(5 1) vs (4 1)	0.11	0.02	4.91	0.00	0.07	0.16
(6 0) vs (4 1)	0.01	0.02	0.40	0.69	-0.04	0.06
(6 1) vs (4 1)	0.11	0.02	4.84	0.00	0.06	0.15
(7 0) vs (4 1)	0.13	0.02	5.23	0.00	0.08	0.17
(7 1) vs (4 1)	0.17	0.02	7.42	0.00	0.12	0.21
(8 0) vs (4 1)	0.19	0.02	8.42	0.00	0.15	0.24
(8 1) vs (4 1)	0.25	0.02	11.02	0.00	0.20	0.29
(9 0) vs (4 1)	0.33	0.02	15.34	0.00	0.28	0.37
(9 1) vs (4 1)	0.32	0.02	13.83	0.00	0.27	0.36
(10 0) vs (4 1)	0.58	0.02	29.65	0.00	0.55	0.62
(10 1) vs (4 1)	0.41	0.03	16.35	0.00	0.36	0.46
(5 1) vs (5 0)	0.13	0.03	4.86	0.00	0.08	0.18
(6 0) vs (5 0)	0.03	0.03	0.92	0.36	-0.03	0.08
(6 1) vs (5 0)	0.12	0.03	4.78	0.00	0.07	0.17
(7 0) vs (5 0)	0.14	0.03	5.17	0.00	0.09	0.20
(7 1) vs (5 0)	0.18	0.03	7.02	0.00	0.13	0.23
(8 0) vs (5 0)	0.21	0.03	7.90	0.00	0.16	0.26
(8 1) vs (5 0)	0.26	0.03	10.11	0.00	0.21	0.31
(9 0) vs (5 0)	0.34	0.03	13.69	0.00	0.29	0.39
(9 1) vs (5 0)	0.33	0.03	12.59	0.00	0.28	0.38
(10 0) vs (5 0)	0.60	0.02	25.34	0.00	0.55	0.65
(10 1) vs (5 0)	0.43	0.03	15.06	0.00	0.37	0.48
(6 0) vs (5 1)	-0.10	0.02	-4.07	0.00	-0.15	-0.05
(6 1) vs (5 1)	-0.00	0.02	-0.21	0.83	-0.05	0.04
(7 0) vs (5 1)	0.02	0.02	0.64	0.52	-0.03	0.06
(7 1) vs (5 1)	0.06	0.02	2.51	0.01	0.01	0.10
(8 0) vs (5 1)	0.08	0.02	3.57	0.00	0.04	0.13
(8 1) vs (5 1)	0.14	0.02	6.08	0.00	0.09	0.18
(9 0) vs (5 1)	0.22	0.02	10.17	0.00	0.17	0.26
(9 1) vs (5 1)	0.21	0.02	9.00	0.00	0.16	0.25
(10 0) vs (5 1)	0.47	0.02	24.14	0.00	0.44	0.51
(10 1) vs (5 1)	0.30	0.03	11.99	0.00	0.25	0.35
(6 1) vs (6 0)	0.10	0.02	3.98	0.00	0.05	0.14
(7 0) vs (6 0)	0.12	0.03	4.43	0.00	0.06	0.17
(7 1) vs (6 0)	0.16	0.02	6.36	0.00	0.11	0.21
(8 0) vs (6 0)	0.18	0.03	7.28	0.00	0.13	0.23
(8 1) vs (6 0)	0.24	0.02	9.62	0.00	0.19	0.28
(9 0) vs (6 0)	0.32	0.02	13.41	0.00	0.27	0.36
(9 1) vs (6 0)	0.31	0.03	12.23	0.00	0.26	0.36
(10 0) vs (6 0)	0.57	0.02	25.88	0.00	0.53	0.62
(10 1) vs (6 0)	0.40	0.03	14.79	0.00	0.35	0.46
(7 0) vs (6 1)	0.02	0.02	0.85	0.40	-0.03	0.07
(7 1) vs (6 1)	0.06	0.02	2.79	0.01	0.02	0.10
(8 0) vs (6 1)	0.09	0.02	3.88	0.00	0.04	0.13
(8 1) vs (6 1)	0.14	0.02	6.47	0.00	0.10	0.18
(9 0) vs (6 1)	0.22	0.02	10.72	0.00	0.18	0.26
(9 1) vs (6 1)	0.21	0.02	9.46	0.00	0.17	0.25
(10 0) vs (6 1)	0.48	0.02	25.30	0.00	0.44	0.52
(10 1) vs (6 1)	0.31	0.02	12.45	0.00	0.26	0.35

(7 1) vs (7 0)	0.04	0.02	1.71	0.09	-0.01	0.09
(8 0) vs (7 0)	0.07	0.02	2.71	0.01	0.02	0.11
(8 1) vs (7 0)	0.12	0.02	5.03	0.00	0.07	0.17
(9 0) vs (7 0)	0.20	0.02	8.75	0.00	0.16	0.24
(9 1) vs (7 0)	0.19	0.02	7.80	0.00	0.14	0.24
(10 0) vs (7 0)	0.46	0.02	21.40	0.00	0.42	0.50
(10 1) vs (7 0)	0.29	0.03	10.77	0.00	0.23	0.34
(8 0) vs (7 1)	0.02	0.02	1.09	0.27	-0.02	0.07
(8 1) vs (7 1)	0.08	0.02	3.55	0.00	0.04	0.12
(9 0) vs (7 1)	0.16	0.02	7.52	0.00	0.12	0.20
(9 1) vs (7 1)	0.15	0.02	6.54	0.00	0.10	0.19
(10 0) vs (7 1)	0.42	0.02	21.32	0.00	0.38	0.46
(10 1) vs (7 1)	0.25	0.03	9.77	0.00	0.20	0.29
(8 1) vs (8 0)	0.05	0.02	2.41	0.02	0.01	0.10
(9 0) vs (8 0)	0.13	0.02	6.26	0.00	0.09	0.18
(9 1) vs (8 0)	0.12	0.02	5.39	0.00	0.08	0.17
(10 0) vs (8 0)	0.39	0.02	19.73	0.00	0.35	0.43
(10 1) vs (8 0)	0.22	0.03	8.69	0.00	0.17	0.27
(9 0) vs (8 1)	0.08	0.02	3.83	0.00	0.04	0.12
(9 1) vs (8 1)	0.07	0.02	3.10	0.00	0.03	0.11
(10 0) vs (8 1)	0.34	0.02	17.49	0.00	0.30	0.38
(10 1) vs (8 1)	0.17	0.02	6.67	0.00	0.12	0.22
(9 1) vs (9 0)	-0.01	0.02	-0.48	0.63	-0.05	0.03
(10 0) vs (9 0)	0.26	0.02	14.27	0.00	0.22	0.29
(10 1) vs (9 0)	0.09	0.02	3.60	0.00	0.04	0.13
(10 0) vs (9 1)	0.27	0.02	13.48	0.00	0.23	0.31
(10 1) vs (9 1)	0.10	0.03	3.81	0.00	0.05	0.15
(10 1) vs (10 0)	-0.17	0.02	-7.59	0.00	-0.22	-0.13

3 Additional Ecological Predictors at Department Level

3.1 PCGDP

There is a positive and quadratic relationship between PCGDP (Per Capita Gross Domestic Product) and SWB in figure 1, but there is a high spread in SWB (y-axis) along quadratic fit. Among departments with low PCGDP of 2-5k USD, there is a large variability in SWB: ranging from least happy Choco at 7.6 to quite happy Vaupes at 8.3. And then at the high-end of PCGDP of 13-15, there is moderately happy Casanare at 8 to very happy Santander at 8.4. There is also large spread horizontally: many departments with the same SWB on y-axis have very different PCGDP levels on x-axis. Still the relationship is positive and quadratic, as expected and as found in the literature–there are diminishing marginal returns from income in SWB–even in moderately poor Colombia, we find.

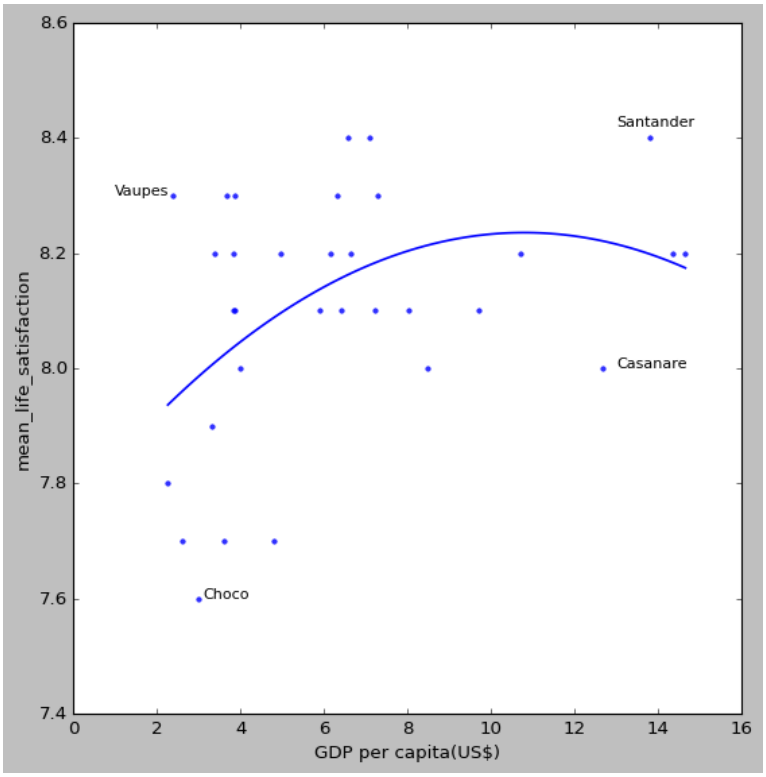


Figure 1

3.2 Population Density

We excluded Bogota because it is not a regular department but capital and San Andres, which is an island. On population density in figure 2 a clear outlier is Atlantico with density at 750,[4] more than double 2nd Quindio at about 300 (not labeled). Most departments are very sparsely populated at below or around 100 people per square kilometer, and there is a great variability in SWB among them from unhappy Choco at 7.6 to several happy departments at 8.4. The denser departments at around 200-300 tend to quite happy at $>$8.1, but some less dense departments are also at $>$8.1. This positive relationship would contradict happy farmer hypothesis, however, departments tend to be large and there is a wide variety of density within, hence the happy farmer hypothesis is not necessarily contradicted.

[4] Atlantico is a relatively small department with large Barranquilla-Soledad metro.

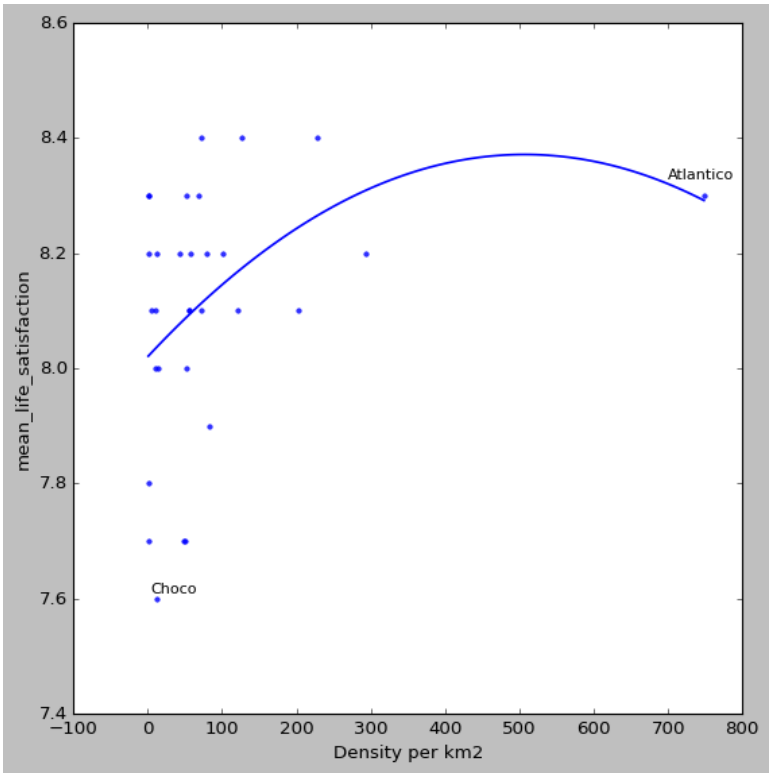


Figure 2

3.3 Crime

The relationship is inverted U-shape in figure 3, opposite to expected negative relationship. However, again, departments tend to be large and there is a wide variety of crime within. And the link between homicide rate in a department and happiness of a person living in that department is much weaker than safety perception of a person and her happiness as studied in the body of the paper.

It is often rather subjective perceptions than objective facts that affect person's wellbeing more.[5]

It may be more of perception of crime as captured by safety satisfaction in the body of the paper than objective crime, which usually disproportionately affects small minority of population such as narcotrafficers and guerrillas. In fact it may be petty crime that upsets an average Colombian more than serious violent crime like homicides. For instance, in Cali while homicide rate has decreased considerably over the years, the petty crime such as petty theft remains high and bothersome for an average Colombian (Martínez and Short 2020).

[5]As the saying goes, "the city on the ground and the city in the mind" (Michael Pacione)–there is a contrast between the objective, physical reality of a city and the subjective, psychological experience of its inhabitants `https://www.sciencedirect.com/science/article/pii/S0169204602002347`.

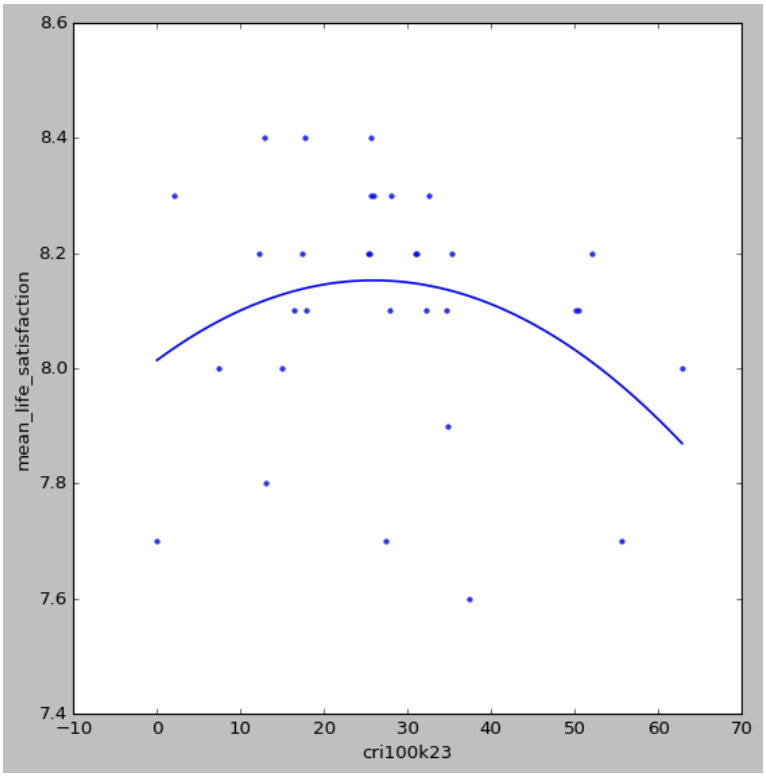


Figure 3

Crime in the West, notably in the US, tends to be urban, but in Colombia in addition to high urban crime, there is also also high rural crime.

Large Colombian cities have its ghettos with rampant crime. And rural geography of crime has to do with infrastructure and climate. Infrastructure: roads used to transport drugs and sea ports used to ship it. At low altitude in warm climate there is coca cultivation, but at high altitudes in cold climate it is more peaceful as one can only farm onions, potatoes, and so forth.

4 Re-estimation Using 2021 ECV

2021 May be biased say because of covid, especially that it peaked in Colombia in 2021 `https://coronavirus.jhu.edu/region/colombia`(by 2023 used in the paper it pretty much cleared). But it a useful robustness check, if covid didn't change the results much, then the results should be pretty stable. Also note that covid in Colombia had quite high mortality `https://coronavirus.jhu.edu/data/mortality`

4.1 Departments/Provinces

Again as said in the paper, overall results are similar to 2023 reported in the body of the paper, except both small sample Amazonas and Vaupés, South-Eastern and with 8.3 in 2023 v 7.6 in 2021, a .7 difference. Second largest differences are for Santander and Putumayo which increased by .4 from 2021 to 2023.

Table 2: Life satisfaction by department.

department	life satisfaction 2023	total obs 2023	non-missing life satisfaction 2023	mean life satisfaction 2021	total obs 2021	non missing life satisfaction 2021	2023-2021
Amazonas	8.3	276	245	7.6	423	365	0.7
Antioquia	8.2	5499	4729	8.3	6036	5060	-0.1
Arauca	8.1	974	773	8.1	1103	868	0
Atlántico	8.3	3144	2311	8.0	2338	1799	0.3
Bogotá, D.C.	8.2	4688	3446	8.1	5337	3810	0.1
Bolívar	8.2	2584	2258	8.1	2495	2156	0.1
Boyacá	8.0	4129	3649	7.8	4638	4122	0.2
Caldas	8.4	2822	2555	8.2	3335	3005	0.2
Caquetá	8.1	2547	2018	7.9	2846	2249	0.2
Casanare	8.0	1749	1370	8.1	2051	1457	-0.1
Cauca	7.7	2580	2244	7.9	2572	2287	-0.2
Cesar	8.3	2224	1717	8.0	1833	1510	0.3
Chocó	7.6	857	754	7.9	1016	913	-0.3
Cundinamarca	8.1	3818	3322	7.8	4233	3684	0.3
Córdoba	8.1	2706	2240	8.0	2226	1933	0.1
Guainía	7.7	327	273	7.4	350	274	0.3
Guaviare	8.2	700	457	8.0	621	411	0.2
Huila	8.1	2961	2322	8.0	2999	2407	0.1
La Guajira	8.2	951	750	7.9	988	736	0.3
Magdalena	8.2	2458	2096	8.0	2110	1710	0.2
Meta	8.2	3683	2752	8.1	3862	2817	0.1
Nariño	7.7	2284	2034	7.4	2847	2492	0.3
Norte de Santander	8.3	2499	2028	8.2	2587	2171	0.1
Putumayo	8.0	950	752	7.6	1272	1012	0.4
Quindío	8.2	2085	1658	8.4	2186	1785	-0.2
Risaralda	8.4	1634	1353	8.3	1996	1663	0.1
San Andrés y Providencia	8.4	28	25	8.2	29	26	0.2
Santander	8.4	3790	3194	8.0	4153	3574	0.4
Sucre	7.9	1798	1399	7.8	1899	1467	0.1
Tolima	8.1	3564	3215	8.0	4257	3770	0.1
Valle del Cauca	8.1	4892	4093	8.1	5268	4421	0
Vaupés	8.3	265	245	7.6	288	267	0.7
Vichada	7.8	515	455	7.6	584	482	0.2

Figure 4 shows 95% CI and hence whether the differences between departments are significant.

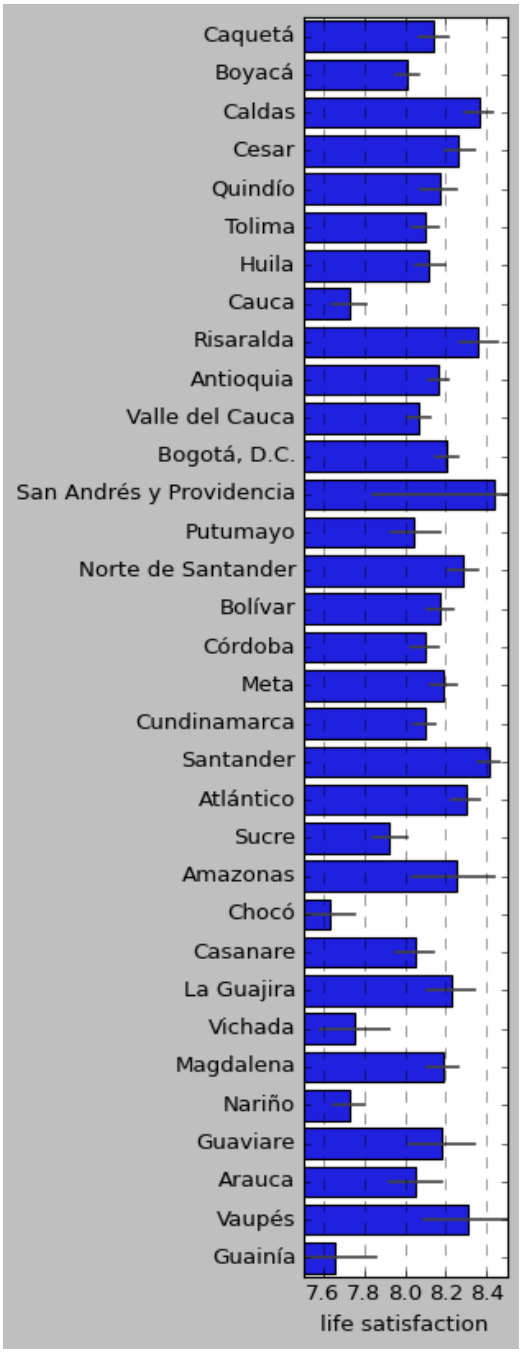


Figure 4: 95% CI shown.

4.2 Cities/Municipalities

This robustness check is very important because ECV is not representative of municipalities.

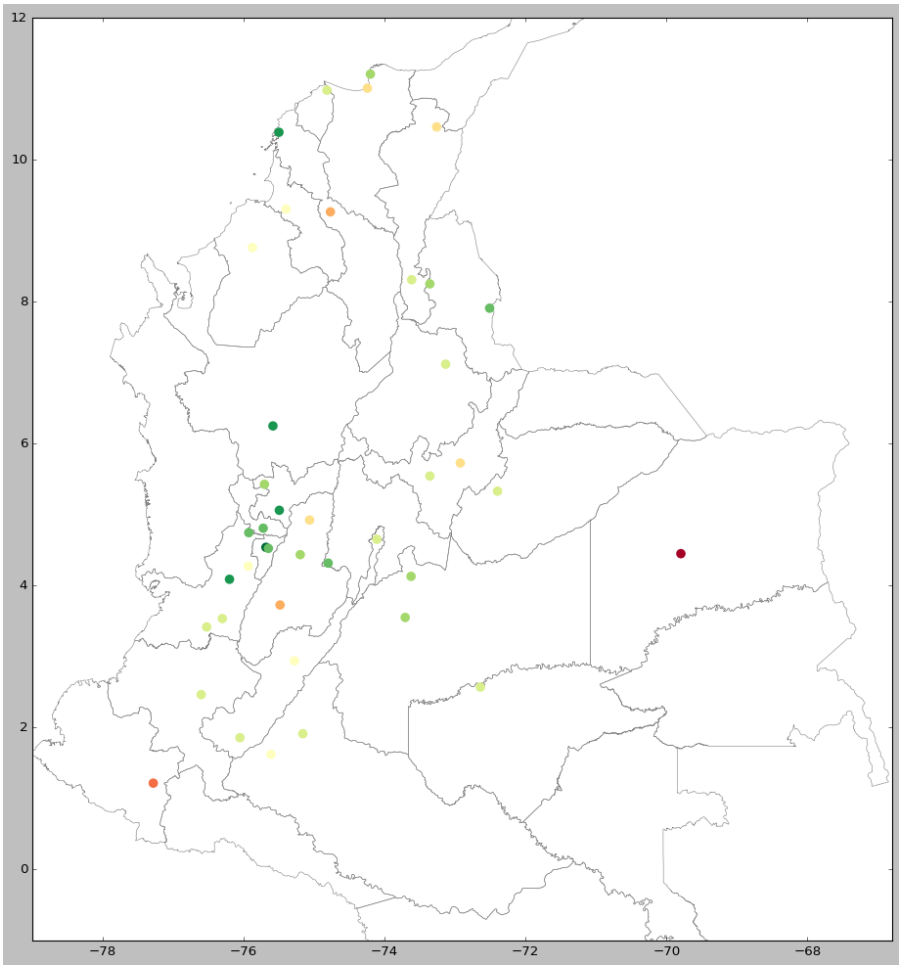


Figure 5: Zoom in to see the detail

Table 3: Life satisfaction by municipality, where non-missing on life satisfaction >200.

municipality name	mean life satisfaction 2021	non missing life satisfaction 2021	mean life satisfaction 2023	non missing life satisfaction 2023	2023-2021 diff
MEDELLÍN	8.4	959.0	8.4	905.0	0.0
BARRANQUILLA	8.1	1116.0	8.3	1363.0	0.2
SABANALARGA	NaN	NaN	8.3	234.0	NaN
BOGOTÁ, D.C.	8.1	3836.0	8.2	3446.0	0.1
CARTAGENA DE INDIAS	8.4	572.0	8.2	591.0	-0.2
EL CARMEN DE BOLÍVAR	NaN	NaN	7.8	200.0	NaN
MAGANGUÉ	7.8	207.0	8.3	205.0	0.5
TUNJA	8.1	241.0	7.9	270.0	-0.2
SOGAMOSO	7.9	367.0	8.2	236.0	0.3
MANIZALES	8.4	567.0	8.4	462.0	0.0
RIOSUCIO	8.2	203.0	8.5	203.0	0.3
FLORENCIA	8.0	630.0	8.2	575.0	0.2
POPAYÁN	8.1	232.0	7.9	225.0	-0.2
VALLEDUPAR	7.9	274.0	8.0	339.0	0.1
AGUACHICA	8.1	215.0	8.6	202.0	0.5
MONTERÍA	8.0	372.0	8.3	465.0	0.3
LORICA	NaN	NaN	8.1	201.0	NaN
GIRARDOT	8.3	222.0	8.1	200.0	-0.2
NEIVA	8.0	436.0	8.3	417.0	0.3
PITALITO	8.1	288.0	8.0	293.0	-0.1
SANTA MARTA	8.2	218.0	8.3	225.0	0.1
CIÉNAGA	7.9	252.0	8.4	234.0	0.5
EL BANCO	NaN	NaN	8.3	200.0	NaN
VILLAVICENCIO	8.2	990.0	8.2	982.0	0.0
GRANADA	8.2	233.0	8.3	229.0	0.1
PASTO	7.7	309.0	7.9	244.0	0.2
SAN JOSÉ DE CÚCUTA	8.3	553.0	8.3	573.0	0.0
ARMENIA	8.5	681.0	8.1	668.0	-0.4
CALARCÁ	8.3	257.0	8.4	244.0	0.1
PEREIRA	8.3	677.0	8.4	524.0	0.1
BUCARAMANGA	8.1	842.0	8.6	798.0	0.5
BARRANCABERMEJA	NaN	NaN	8.6	270.0	NaN
SINCELEJO	8.0	275.0	8.1	222.0	0.1
IBAGUÉ	8.2	549.0	8.4	478.0	0.2
SANTIAGO DE CALI	8.1	1108.0	8.0	985.0	-0.1
CARTAGO	8.3	279.0	8.2	220.0	-0.1
PALMIRA	8.1	332.0	7.9	319.0	-0.2
TULUÁ	8.4	307.0	8.2	303.0	-0.2
TAME	NaN	NaN	8.0	203.0	NaN
YOPAL	8.1	240.0	8.3	264.0	0.2
SAN JOSÉ DEL GUAVIARE	8.1	231.0	8.1	246.0	0.0
CUMARIBO	7.5	215.0	7.5	225.0	0.0

Figure 6 shows 95% CI and hence whether the differences between departments are significant.

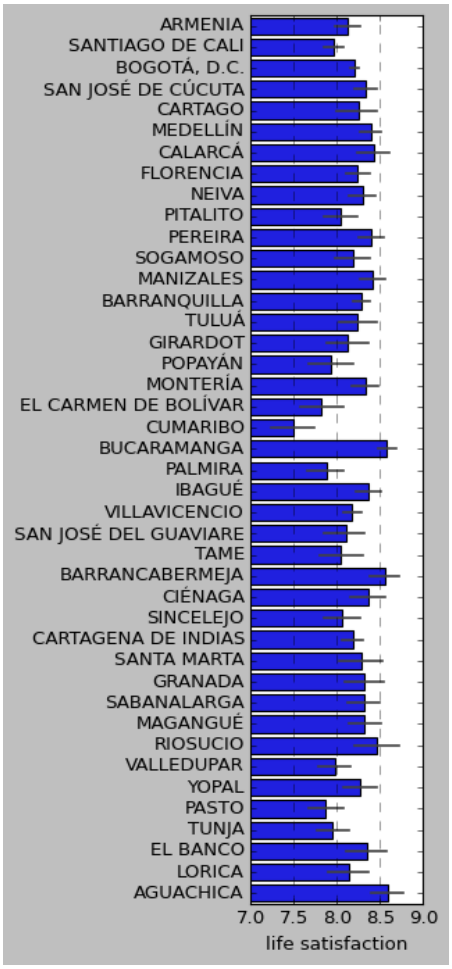


Figure 6: 95% CI shown.

4.3 Urban-Rural and Farmers

Colombia is quite urbanized at around 70-80%. `https://population.un.org/wup/countryprofiles?country=Colombia&mainCategory`
`Countries%20and%20Aggregates&subCategory=Urbanization%20-%20International%20definition`

Still, a substantial part at 20-30% is rural.

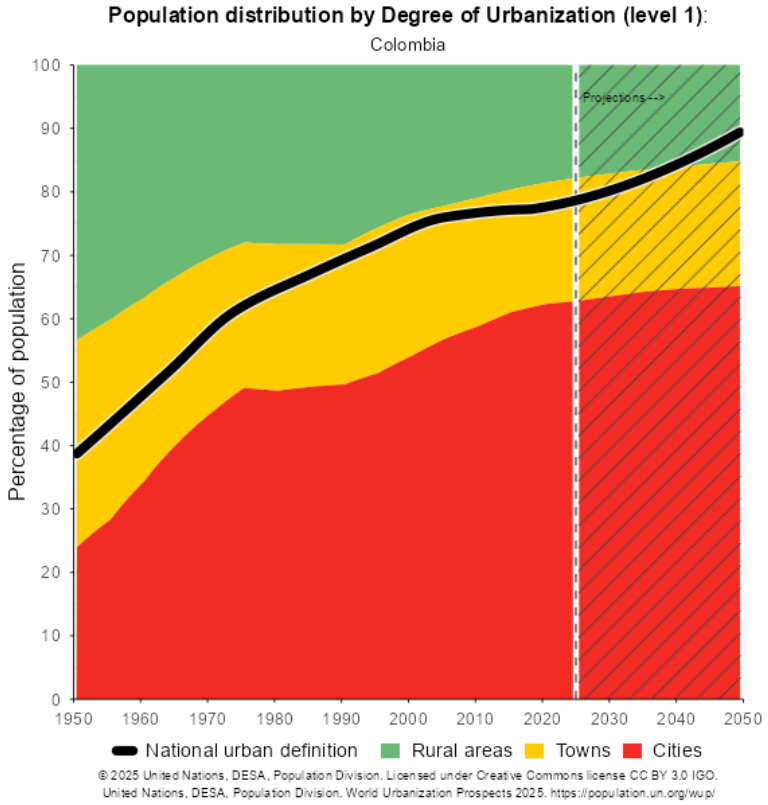


Figure 7

For rural in 2021 pattern is similar to 2023 in body of the paper but less strong for lower deciles where rural are happier than urban, but stronger for 2 top deciles where the opposite is true. For farmers in 2021 there is no happiness premium in lower deciles, but for non-farmers there is happiness premium in 2 top deciles.

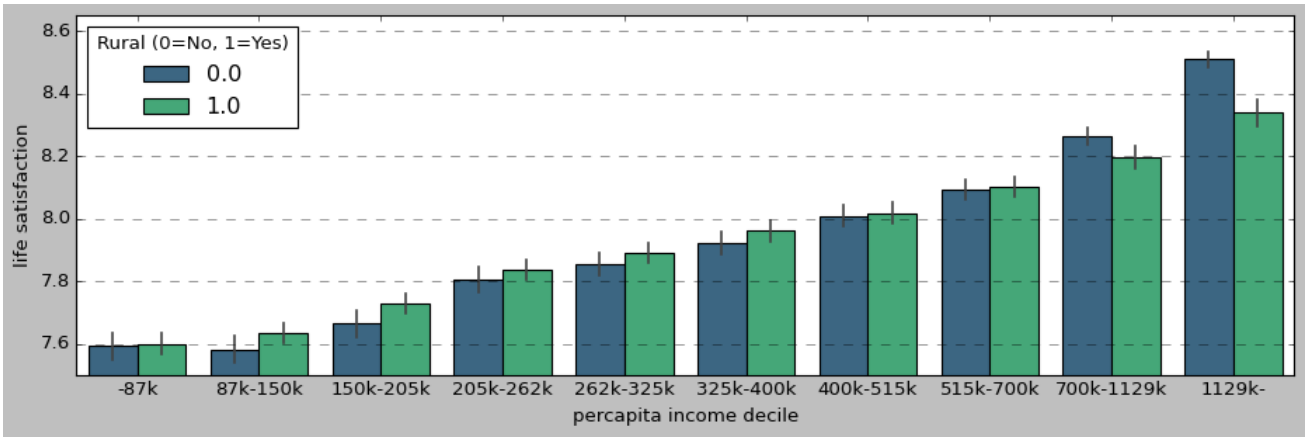


Figure 8

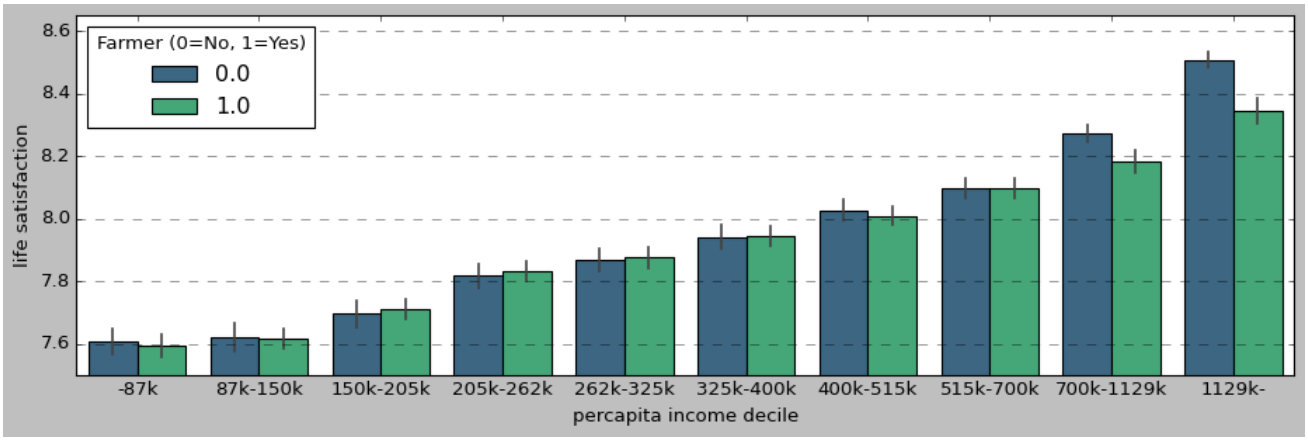


Figure 9

5 Geography of Colombia

5.1 Administrative Boundaries

Colombia is at North-West of South America (fig 10), a large country of about 54m, third largest in Latin America after Brazil and Mexico (`https://fred.stlouisfed.org/series/POPTOTCOA647NWDB`.) Colombia is also large in land mass at about 1.1m sq km, while much smaller than Brazil at 8.5m sq km, it is larger than Venezuela at .9m sq km, and about 3x larger than Germany at .35m sq km and about 2x of Spain at .5 sq km.



Figure 10: Colombia circled in red in Latin America shaded in green.

Map of the 32 departments is in figure 11.



Figure 11: Colombia 32 departments. Source: `https://upload.wikimedia.org/wikipedia/commons/thumb/6/67/Departments_of_colombia.svg/960px-Departments_of_colombia.svg.png`.

5.2 Physical Geography

The paper is about happiness geography, but it would be incomplete without physical geography. Topography is well mapped and best viewed at `https://en.wikipedia.org/wiki/File:Mapa_de_Colombia_(relieve).svg`, where you can zoom in to see the detail. Population is nicely visualized in a 3d map at `https://geoportal.dane.gov.co/geovisores/sociedad/grilla-dane-vihope/`. To compare topography v population density here side by side see figure 12. Much of the population is concentrated around mountainous areas (with a notable exception of quite populated and largely flat Caribbean region in North).

Colombia's road infrastructure and connectivity is very poor.[6] Then this could be one explanation for large distinctiveness/uniqueness of regions–they have not become homogeneous as in many other countries as they are weakly connected due to mountains and poor road infrastructure.

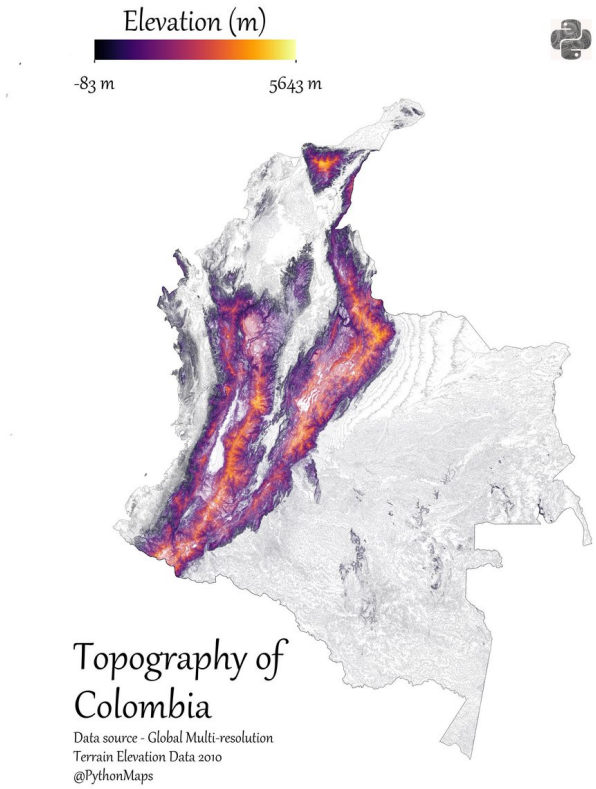
Virtually all of Eastern Colombia, about half of the country, is uninhabited road and city free pristine nature stretching from Eastern Arauca and Vichada down to Amazonas. Likewise, but much narrower, pristine nature belt stretches over much of Western Pacific, especially in Choco.

Colombian population is concentrated along 2 valleys in Andes–one along Panamericana highway from second largest Medellin through Cafetero region, third largest Cali, down to Ipiales on border with Ecuador. The second valley runs from Tunja to largest Bogota down to Neiva. And there is also quite populated Caribbean.

[6] 2018 quality of roads rank is 110/137 (`https://reports.weforum.org/pdf/gci-2017-2018-scorecard/WEF_GCI_2017_2018_Scorecard_EOSQ057.pdf`)–part of the problem is mountains.

Figure 12: Topography and population density. Both maps come from visually appealing PythonMaps: hi-res versions are at the below links. A 3d topography map is also at `https://www.reddit.com/r/MapPorn/comments/u9gr9j/the_topography_of_colombia`. And another topography map at `https://upload.wikimedia.org/wikipedia/commons/3/3b/Colombia_Topography.png`.

(a) Topography `https://x.com/PythonMaps/status/1663937378782937088/photo/1`



(b) Population density `https://x.com/PythonMaps/status/1699089505133772803/photo/1`



Last but not least ecological footprint. `https://data.footprintnetwork.org/` is a useful global overview, Colombian footprint per person is quite low at 2gha, 4x lower than that of the US at 8 gha, and it is stable since 1960s, but bio-capacity per person has declined from 11.4 gha in 1960 to 3.3 in 2024. `https://happyplanetindex.org/countries/COL/` is another tool for cross-country comparisons.

Figure 13 shows useful hi-resolution visualizations of footprint across Colombia for 2015.[7]

[7]OECD provides more recent update across Colombia: `https://www.oecd.org/en/publications/oecd-environmental-performance-reviews-colombia-2026_968398f7-en/full-report/biodiversity-conservation-and-sustainable-use_2099ac2c.html`

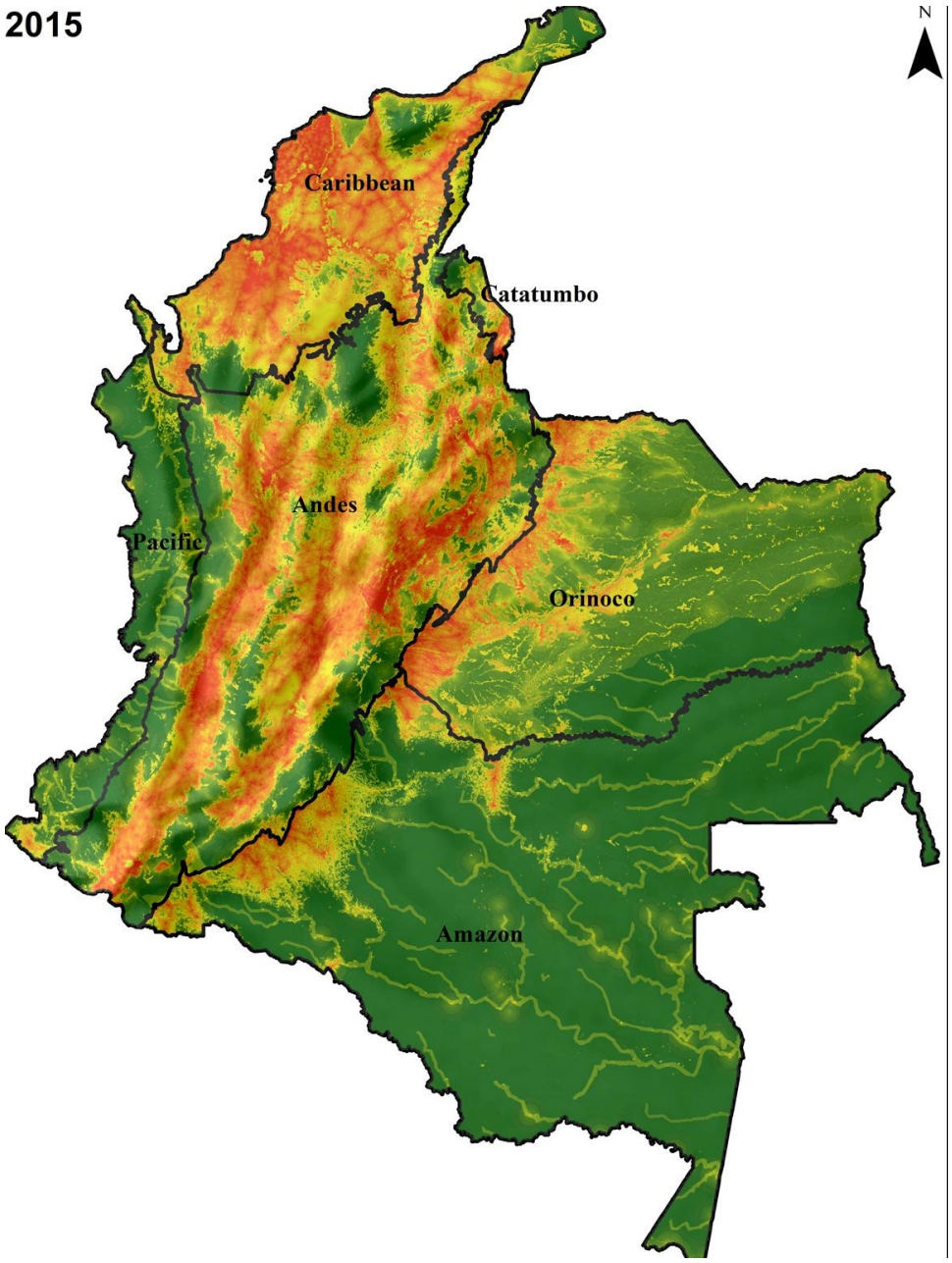
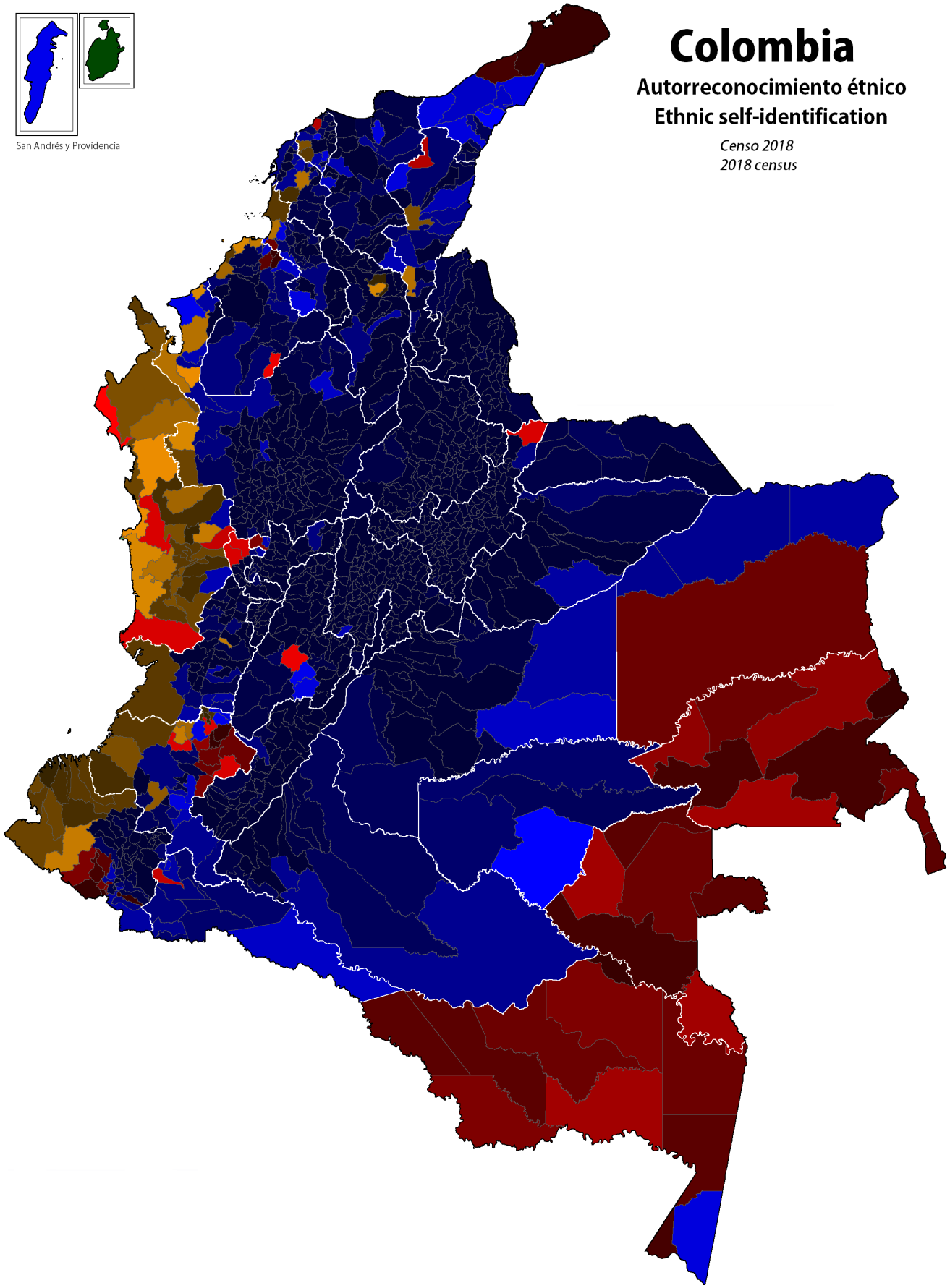


Figure 13: Human footprint: Low-Medium-High: Green-Yellow-Red. Source: `https://www.sciencedirect.com/science/article/pii/S1470160X20305677`

5.3 Socio-cultural Geography

Ethnic self-identification is shown in figure 14.


Colombia
Autorreconocimiento étnico
Ethnic self-identification
Censo 2018
2018 census
San Andrés y Providencia
>30% >35% >40% >45% >50% >55% >60% >65% >70% >75% >80% >85% >90% >95%
Mestizos y blancos
Afrocolombiano
Indígena
Raizal
Gaël Malo L'Hermine
@glhermine
Ottawa, ON (2020)

Colombia has extraordinary diversity, not just biodiversity (2nd largest in the world after 8x larger Brazil), but also cultural and social. Colombia can be thought of as a microcosm of Latin America culturally and socially[8] with Indigenous, African, and Spanish heritage. Indigenous heritage and culture is still alive and widespread, as opposed to the US where it was largely eliminated (Whittle 2026).

Administratively Colombia has 32 departments further divided into municipalities. They are very diverse, almost as if different countries. And there are large differences within departments and across their municipalities as well. For instance in Narino, Pasto is very different from Tumaco. Likewise, in Valle de Cauca, Cali is very different from Buenaventura.

Urbanization differs widely–and urbanization is not just built environment by way of life (Wirth 1938). Again about half of eastern Colombia is wilderness, then there are many small towns and villages with slow and communal way of life, and then there are also a handful of cities where way of life is more Western-like.

For instance, Bogota (or at least parts of it) is similar to other large fast paced commerce oriented cities like New York or London. Likewise, it is very international and has the most connected airport in Latin America (most international connections) and one of the 3 largest airports in Latin America (close to Sao Paulo and Mexico city).[9] Cartagena is Miami-like: expensive and fashionable catering to the rich and tourists. On the other hand, Cali, the third largest city, is very much Latin with more laid back atmosphere and salsa music. Indeed, one could think these cities belong to different countries or even cultures. Pacific Buenaventura and Tumaco are different yet–less organized and impoverished. One could enumerate many more examples of differences across municipalities.

Bogota could be considered a primate city, although the difference versus second largest Medellin is only of order of about 2, smaller than in many other countries[10]

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[8] See https://colombia.co/en/colombia-country/history/greatest-cultural-blend and Okulicz-Kozaryn and Martinez (2025). Also physically–the only South American country with both Pacific and Atlantic coasts, and has just about any type of topography, climate, and vegetation: Andes with several snowy peaks over 5km, hot and dry deserts, humid rain forests, and moderate climate valleys.

[9] https://www.oag.com/blog/latin-americas-most-connected-airports-aviation-infographics and https://en.wikipedia.org/wiki/List_of_the_busiest_airports_in_Latin_America.

[10] https://en.wikipedia.org/wiki/Primate_city.

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